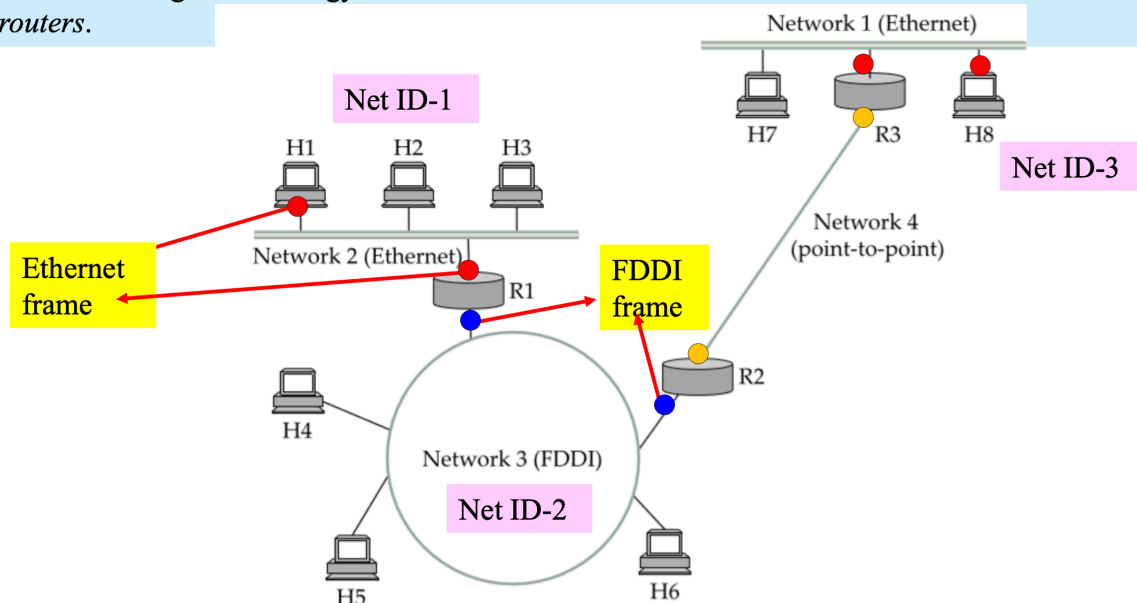


List of question of slide - 3

1. Give an example of heterogenous network with activities of routers. (P: 3-4) Diagram P-4 only)

Figure below shows an example *internetwork*. An *internetwork* is often referred to as a **network of networks** because it is made up of lots of smaller networks. In this figure, we see Ethernets, an FDDI (Fibre Distributed Data Interface) ring, and a point-to-point link. Each of these is a single technology network. The nodes that interconnect the networks are called *routers*.



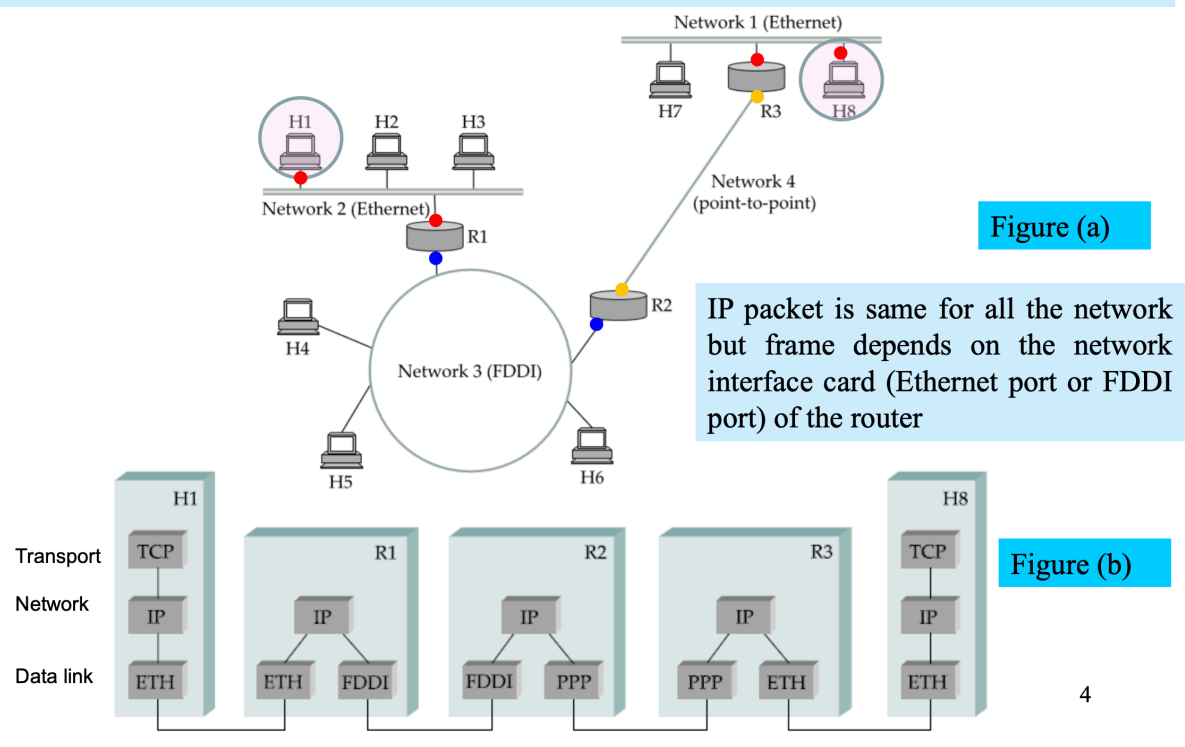
A simple internetwork: Hn = host; Rn = router

Router(config)# interface fddi 4/0

Router(config)# interface fa 4/0

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Figure below shows how hosts H1 and H8 are logically connected by the internet in including the protocol graph running on each node. Let H1 transmits a TCP packet to node H8.



2. What is the function of different services of the ip header (p-7 not fig)

The **Differentiated Services (DS) field** is an 8-bit field in the IP header used to specify how packets should be handled based on the type of service they require. It allows routers and switches to prioritize certain types of traffic to ensure **Quality of Service (QoS)**.

It is intended to distinguish between different classes of service. Various combinations of reliability and speed are possible.

- For digitized voice, fast delivery is more essential than accurate delivery.
- For digitized video, delay is tolerable but prefers smooth playback.
- For file transfer, error-free transmission is more important than fast transmission.

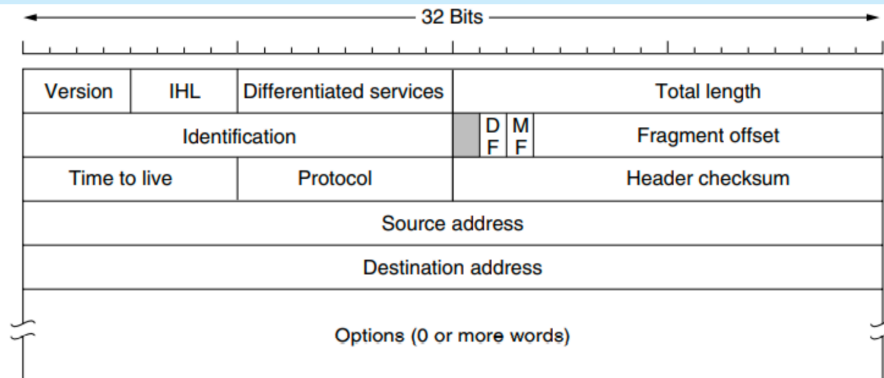
This field is related to three parameters: {Delay, Throughput, Reliability} in communication.

- **Delay:** How quickly data must reach its destination.
- **Throughput:** How much data must be delivered in a certain amount of time.
- **Reliability:** How accurate or error-free the transmission must be.

3. TTL (p: 12)

Time-To-Live (8 bits)

The time-to-live (TTL) field is used to control the maximum number of hops (routers) visited by the datagram. When a source host sends the datagram, it stores a number in this field. This value is approximately 2 times the maximum number of routes between any two hosts. Each router that processes the datagram decrements this number by one. If this value, after being decremented, is zero, the router discards the datagram.



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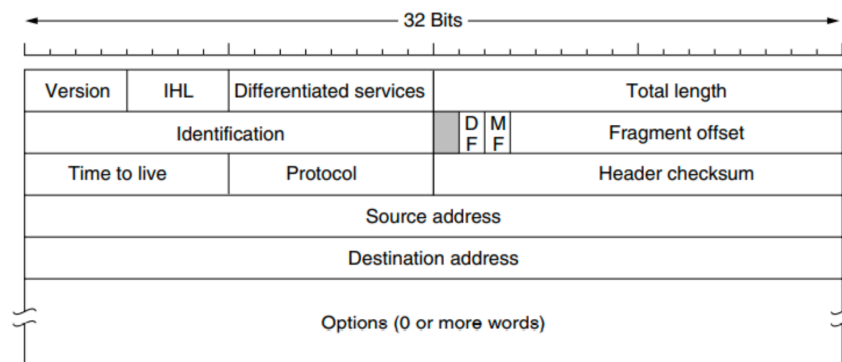
4. Give an example of calculation of header checksum (p: 15-17)

Example: Calculating a checksum

❑ The header is shown in red and the checksum is underlined.
4500 0073 0000 4000 4011 b861 c0a8 0001 c0a8 00c7 0035 e97c 005f 279f 1e4b 8180

❑ To calculate the checksum, we can first calculate the sum of each 16 bit value within the header, skipping only the checksum field itself. Note that the values are in hexadecimal notation.

4500 + 0073 + 0000 + 4000 + 4011 + c0a8 + 0001 + c0a8 + 00c7
= 2479C (equivalent to 149,404 in decimal).



❑ Convert the value 2479C to binary:

0010 0100 0111 1001 1100

The first 4 bits are the carry and will be added to the rest of the value:

0010 + 0100 0111 1001 1100 = 0100 0111 1001 1110

Next, we flip every bit (1's complement) in that value, to obtain the checksum:

0100 0111 1001 1110 becomes:

1011 1000 0110 0001

This is equal to **B861** in hexadecimal, as shown underlined in the original IP packet header.

❑ Verifying a checksum

When verifying a checksum, the same procedure is used as above, except that the original header checksum is not omitted.

$4500 + 0073 + 0000 + 4000 + 4011 + \text{b861} + \text{c0a8} + 0001 + \text{c0a8} + 00c7 = 2fffd$

Add the carry bits:

$fffd + 2 = ffff$

Taking the 1's complement (flipping every bit) yields 0000, which indicates that no error is detected. IP header checksum does not check for the correct order of 16 bit values within the header.

5. Give an example of tunnelling of IP packet (p: 23-24, fig ektai)

Example: IPv6-over-IPv4 Tunneling

Let's say you have:

- A source computer A that only supports **IPv6**.
- A destination computer F that also supports **IPv6**.
- But the **intermediate network (like the internet, C,D)** only supports **IPv4**.

How tunneling works here:

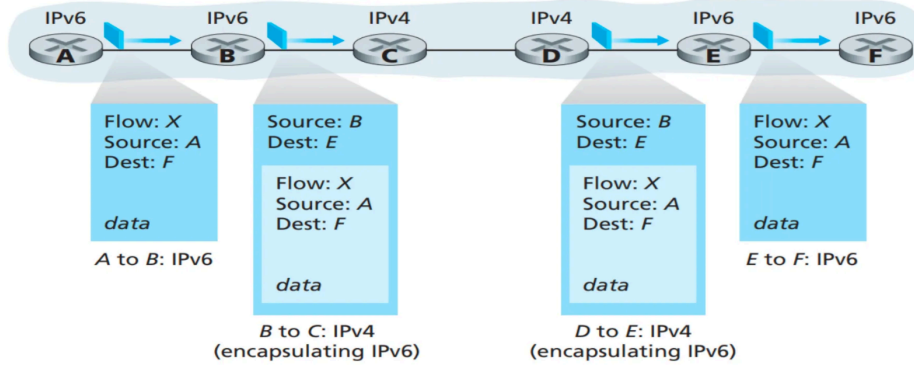
1. The source computer A creates an **IPv6 packet** intended(উদ্দেশ্য) for the destination.
2. This IPv6 packet is **encapsulated** inside an **IPv4 packet** at the source gateway B.
3. Then the encapsulated packet travels through the **IPv4-only internet C→D**.
4. At the destination gateway E, the **IPv4 header is removed**, revealing the original IPv6 packet.

5. Then the IPv6 packet is then forwarded to the destination computer F.

Logical view



Physical view



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